

# Paavan Randhawa

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## Profile

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Third-year BSc student in Mathematics and Computing Science at Simon Fraser University (GPA 3.57, Dean's Honour Roll), with hands-on experience developing software through requirements gathering, technical specification, implementation, testing, code review, and delivery. Experienced in Python, Java, C, C++, JavaScript, and web development through academic projects and software development experience. Comfortable working independently and collaboratively, communicating technical information clearly, and adapting to changing requirements. Expected to graduate April 2028.

## Technical Skills

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**Programming:** Python · Java · C · C++ · JavaScript · SQL · Bash

**Web & Technology:** HTML · CSS · Bootstrap · Tailwind CSS

**Development Tools:** Git · GitHub · Unix/Linux · CMake · Maven · JUnit · JaCoCo · Javadoc · Visual Studio Code

**Methods & Practices:** Software Development Lifecycle (SDLC) · Agile (Scrum) · UML · Test-Driven Development · Code Review

**Productivity:** Microsoft Office (Word · Excel · PowerPoint · Outlook)

**Languages:** English · French (DEL F A2) · Punjabi

## Projects

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**Unix Shell** | C, Unix/Linux, CMake

Jan 2026 – Feb 2026

- Independently developed a Unix-style shell in C, implementing process creation and execution, foreground and background process management, signal handling, built-in commands, and command history within a Unix/Linux environment.
- Planned and implemented components against technical specifications, performed integration testing throughout development, and configured repeatable project builds using CMake.

**Rubik's Cube Solver** | Java

Nov 2025 – Dec 2025

- Developed a Java application that parses Rubik's Cube states and generates step-by-step solutions using the two-phase Kociemba algorithm, translating requirements into a complete working solution.
- Tested the application across a wide range of configurations, investigated incorrect outputs, and refined the implementation to improve reliability.

**Mythos Seafarer** | Java, JUnit, Agile (Scrum)

Jan 2026 – Apr 2026

- Collaborated with a team to develop an arcade-style Java game using Agile (Scrum), contributing collision detection, scoring logic, and multi-level progression to a shared codebase.
- Participated in weekly code reviews, wrote JUnit tests, tracked and communicated development progress and blockers, and refactored features iteratively as requirements evolved across sprints.

**Pocoloco** | Python

Dec 2024

- Designed and developed a multiplayer dice game in Python, implementing randomized turn order, chip-based gameplay, custom scoring combinations, and round-based game state management.
- Independently identified and implemented the program logic required to manage game states and determine outcomes, strengthening structured problem-solving and software design skills.

## Work Experience

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**Scorekeeper**

Apr 2023 – Jul 2024

*Surrey Minor Ball Hockey*

*Surrey, BC*

- Maintained accurate records of scores, penalties, and game timing while coordinating with officials under time pressure.

**Library & Classroom Assistant**

Nov 2022 – Dec 2022

*North Ridge Elementary School*

*Surrey, BC*

- Supported students with reading and math activities, adapting explanations to individual learning needs.
- Helped organise and run school activities, including a book fair and field trips.

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| <b>Retail Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Oct 2022 – Nov 2022 |
| <i>Shoppers Drug Mart</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Surrey, BC</i>   |
| <ul style="list-style-type: none"> <li>– Assisted customers across pharmacy, food, and personal care departments while maintaining organised store operations.</li> <li>– Performed expiry checks, restocked inventory, and supported seasonal merchandising.</li> </ul>                                                                                                                                                                                                                                                   |                     |
| <b>Software Development Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Jun 2022 – Sep 2022 |
| <i>CodeWizardsHQ</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>Remote</i>       |
| <ul style="list-style-type: none"> <li>– Gathered business requirements from a nonprofit client and translated them into technical specifications for full-stack applications using Python, HTML, CSS, and JavaScript.</li> <li>– Developed a password generator with randomized generation and input validation, then tested and reviewed the implementation with peers.</li> <li>– Coordinated with other interns to incorporate feedback, adapt to changing requirements, and deliver solutions on schedule.</li> </ul> |                     |

## Education

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| <b>Simon Fraser University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Burnaby, BC                           |
| <i>BSc, Mathematics and Computing Science · GPA 3.57 · Dean's Honour Roll</i>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Sep 2024 – Apr 2028 (Expected)</i> |
| <ul style="list-style-type: none"> <li>– Shein &amp; Aymnmohamed Rajan Family Entrance Scholarship in Computing Science (Sep 2024)</li> <li>– Relevant coursework: Data Structures &amp; Programming · Software Engineering · Systems Programming · Introduction to Computer Systems · Discrete Mathematics I &amp; II (in progress) · Data Communications &amp; Networking (in progress) · Introduction to Artificial Intelligence (in progress) · Probability &amp; Statistics · Applied Linear Algebra</li> </ul> |                                       |

## Awards & Certifications

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|---------------------------------------------------------------------------------------------|-------------|
| <b>Claude Academy: Claude Code 101</b>   <i>Anthropic</i>                                   | Sep 2026    |
| <b>Schulich Leader Nominee</b>   <i>National STEM leadership recognition</i>                | Feb 2024    |
| <b>BC Achievement Scholarship</b>   <i>Provincial academic excellence award</i>             | Jun 2024    |
| <b>District Scholarship – Excellence in Applied Sciences</b>                                | Jun 2024    |
| <b>Surrey Football Club Scholarship</b>   <i>Athletic and academic recognition</i>          | Jun 2024    |
| <b>CodeWizardsHQ Full-Stack Coding Program</b>   <i>Python, JavaScript, Flask, SQL, Git</i> | 2019 – 2023 |
| <b>DELTA A2</b>   <i>French language proficiency certification, Government of France</i>    | 2024        |

## Volunteering

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| <b>Notetaker – Centre for Accessible Learning</b>                                                                                                                                                                                                   | Sep 2025 – Dec 2025 |
| <i>Simon Fraser University</i>                                                                                                                                                                                                                      | <i>Burnaby, BC</i>  |
| <ul style="list-style-type: none"> <li>– Produced clear, well-organised lecture notes each week for students registered with the Centre, accurately capturing course content and consistently meeting deadlines throughout the semester.</li> </ul> |                     |
| <b>Peer Tutor – Homework Club</b>                                                                                                                                                                                                                   | Sep 2023 – Jun 2024 |
| <i>Panorama Ridge Secondary School</i>                                                                                                                                                                                                              | <i>Surrey, BC</i>   |
| <ul style="list-style-type: none"> <li>– Provided weekly Mathematics and Science tutoring to students, adapting explanations to individual needs and communicating concepts clearly.</li> </ul>                                                     |                     |

## Clubs & Activities

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| <b>Student Organisations:</b> Women in Computing Science (WiCS) · Computer Science Student Society (CSSS) · Mathematics Student Union (MSU)                                                             |  |
| <b>Athletics:</b> Competitive soccer – Surrey United Soccer Club (14+ years) · Competitive ball hockey – Surrey Minor Ball Hockey (since 2021) · Long-distance running (Sun Run 10K) · Hiking · Cycling |  |